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Course: CISP-357 - Introduction to Data Science

Final Project Report

Project Description:

In this project, Open-Meteo API is used to collect real world hourly temperature data for the city of Sacramento. The data represents temperature readings over a seven-day period. The purpose of this project is to demonstrate how to retrieve data from an external API, Parse the data using python, visualize it using a scatter plot and generate a PDF report programmatically. Some of the libraries i used in this program includes reportlab(For generating PDF and to write the report),requests(for requesting the API access),Datetime and Matplotlib(For plotting)

Data Trend Analysis:

The scatter plot represents hourly recorded temperature readings of seven days of weather data from 08/27/2026 through 09/02/2026. The data shows a consistent daily temperature pattern, with higher temperatures during the daytime hours and decrease during the night time hours repeating the cycle across all seven days. During this time period, the highest recorded temperature was 95.4°F, while the lowest temperature was 57.6°F, indicating mild spring weather conditions & The average recorded temperature of the data reflects typical daily temperature variations for Sacramento during this season.

Hourly Temperature Scatter Plot (Sacramento, 7 Days)

